James Mann

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Education

University College London | **Computer Science MEng** | **Sept 2024 – May 2027**

- First year average: 81%
- First Year Representative of the AI Society

Experience

Arcadia Impact | **Research Analyst / Software Engineer** | **May 2025 -**

- Responsible for researching and developing an automated pipeline for implementing and verifying evaluations, to massively accelerate the addition of new benchmarks to Inspect Evals.
- Responsible for researching and implementing
- Led the creation of a technical tender responding to a €1,000,000 tender for AI Evaluations Infrastructure for the EU, ultimately coming second out of fourteen applying consortiums.
- Part of the maintenance team for Inspect Evals, responding to issues and applying QA to prospective new Evaluations.

MATS | **Neel Nanda's MATS Training Phase** | **Sep 2025 - Oct 2025**

- Worked on various mini-projects including an LLM progress bar based on model internals and a project investigating LLM's latent representations of user characteristics (age, sex, star sign).
- Developed a final project investigating model tendencies to reward hack, identifying that a deception direction could control gpt-oss-20b's reward hacking.
- Trained linear probes to catch reward hacking intent. Reconstructed the found directions with an SAE to surface hypotheses on causes.

Bluedot Impact | **Incubator Week + Hackathon Prize** | **Nov 2025**

- Accepted into v1 of Bluedot's Incubator Week
- Spent a week developing a plan to create tailored probes for inference-time monitoring.
- Came 2nd overall, winnning £1000 in Bluedot's 100+ person hackathon solo developing an MVP, creating CUDA kernels to minimise latency of monitoring, eventually optimising it down to <5%.

Impact Research Groups | **Technical AI Safety Research Sprint** | **Feb 2025 - April 2025**

- First author of a paper produced as part of an 8 week competitive research sprint, coming third.
- Identified a safety-critical latent factor in high-dimensional model activation space that could steer between refusal and compliance.

Projects

Eventual-Importance Probing

- Ideated and implemented a novel approach to probe training by trying to predict the probability of a given behaviour following a given token. This provides continuous, maximally granular labels that capture intent.
- Inspired by the work of thought anchors and constitutional classifiers, which shows that LLM non-determinism makes binary labels ineffective.
- Engineered a general open source repository that abstracts training probes on any dataset that can be marked.

Probabilistic Image Generation

- Implemented a denoising diffusion probabilistic model in Pytorch, trained on CIFAR100 to generate novel images from the distribution, iteratively optimising the hyperparameters to improve the plausibility of samples.